

Assignment-5.4

Batch-05

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Task 1: Collecting User Data Securely

Prompt Given to GitHub Copilot

“Generate a Python script to collect user name, age, and email. Add comments on how to anonymize or protect this data.”

Code:

```
assignment-5.py > ...
1  import hashlib
2
3  def collect_user_data():
4      name = input("Enter your name: ")
5      age = int(input("Enter your age: "))
6      email = input("Enter your email: ")
7
8      # Hashing email to anonymize sensitive data
9      hashed_email = hashlib.sha256(email.encode()).hexdigest()
10
11     user_data = {
12         "name": name,
13         "age": age,
14         "email_hash": hashed_email
15     }
16
17     return user_data
18
19 data = collect_user_data()
20 print("User data collected securely.")
21 print(data)
22
```

Output:



```
PS C:\Users\kamul\OneDrive\Desktop\AI ASSISTANT CODING> & C:/Users/kamul/AppData/Local/Programs/Python/Python313/python.exe "c:/Users/kamul/OneDrive/Desktop/AI ASSISTANT CODING/assignment-5.py"
Enter your name: amulya
Enter your age: 20
Enter your email: k.amulya2217@gmail.com
User data collected securely.
{'name': 'amulya', 'age': 20, 'email_hash': 'e31f7dcbb8354e699cf60ae747c4f28bac6710aa9247c0b9fe99c4aec0ec76c9'}
PS C:\Users\kamul\OneDrive\Desktop\AI ASSISTANT CODING>
```

Ethical & Security Explanation:

- Sensitive data like email is hashed instead of stored in plain text.
- Passwords or raw personal data should never be logged or stored unencrypted.
- User consent should always be obtained before collecting personal data.
- This approach follows data minimization and privacy-by-design principles.

Task 2: Sentiment Analysis with Bias Awareness

Prompt Given to GitHub Copilot

“Generate a Python function for sentiment analysis and handle potential biases in the data.”

Code:

```
assignment-5.py > analyze_sentiment
1 def analyze_sentiment(text):
2     # Normalize text to avoid case-based bias
3     text = text.lower()
4     # Remove potentially offensive or biased words
5     biased_words = ["hate", "stupid"]
6     for word in biased_words:
7         text = text.replace(word, "")
8     # Simple sentiment word lists
9     positive_words = ["good", "great", "excellent", "happy", "love"]
10    negative_words = ["bad", "poor", "worst", "angry", "dislike"]
11    score = 0
12    for word in positive_words:
13        if word in text:
14            score += 1
15    for word in negative_words:
16        if word in text:
17            score -= 1
18    return score
19    sentence = "I hate this product"
20    score = analyze_sentiment(sentence)
21    print("Sentiment score:", score)
22
```

Output:

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
Python + v

PS C:\Users\kamul\OneDrive\Desktop\AI ASSISTANT CODING> & C:/Users/kamul/AppData/Local/Programs/Python/Python313/python.exe
Drive/Desktop/AI ASSISTANT CODING/assignment-5.py"
Sentiment score: 0
PS C:\Users\kamul\OneDrive\Desktop\AI ASSISTANT CODING>
```

Ethical Considerations

- Removing offensive or biased words helps reduce **discriminatory outcomes**.
- Sentiment models can reflect **training data bias**.
- Dataset balancing and diverse training data are essential.
- AI outputs should always be **reviewed by humans**.

Task 3: Ethical Product Recommendation System

Prompt Given to GitHub Copilot

“Generate a Python program that recommends products ethically with transparency and fairness.”

Code:

```
assignment-5.py > ...
1  def recommend_products(user_history, products):
2      recommendations = []
3      for product in products:
4          # Avoid favoritism by recommending based on relevance, not brand bias
5          if product["category"] in user_history:
6              recommendations.append(product["name"])
7          # Transparency: explain why recommendations are shown
8          print("Recommendations are based on your past interests.")
9      return recommendations
10 user_history = ["books", "electronics"]
11 products = [
12     {"name": "Python Book", "category": "books"},
13     {"name": "Laptop", "category": "electronics"},
14     {"name": "Shoes", "category": "fashion"}
15 ]
16 print(recommend_products(user_history, products))
17
```

Output:

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
PS C:\Users\kamul\OneDrive\Desktop\AI ASSISTANT CODING> & C:/Users/kamul/AppData/Local/Programs/Python/Python313/python.exe "c:/Users/kamul/OneDrive/Desktop/AI ASSISTANT CODING/assignment-5.py"
Recommendations are based on your past interests.
['Python Book', 'Laptop']
PS C:\Users\kamul\OneDrive\Desktop\AI ASSISTANT CODING>
```

Ethical Explanation:

- Avoids unfair promotion of specific brands.
- Provides **transparency** by explaining recommendation logic.
- Allows future addition of **user feedback and opt-out options**.
- Encourages fairness and accountability in AI decisions.

Prompt Given to GitHub Copilot

Code:

Output:

Ethical Logging Practices

- Logs exclude sensitive personal information.
- Logging is used only for **monitoring and debugging**, not surveillance.
- Complies with **ethical data handling and security standards**.
- Prevents data leakage through logs.

Task 5: Responsible Machine Learning Model

Prompt Given to GitHub Copilot

“Generate a machine learning model and add documentation for responsible usage.”

Code:

```
assignment-5.py > ...
1  # Simple Linear Regression without external libraries
2  x = [1, 2, 3, 4]
3  y = [2, 4, 6, 8]
4  # Calculate means
5  mean_x = sum(x) / len(x)
6  mean_y = sum(y) / len(y)
7  # Calculate slope (m)
8  num = sum((x[i] - mean_x) * (y[i] - mean_y) for i in range(len(x)))
9  den = sum((x[i] - mean_x) ** 2 for i in range(len(x)))
10 m = num / den
11 # Calculate intercept (b)
12 b = mean_y - m * mean_x
13 # Prediction
14 x_new = 5
15 y_pred = m * x_new + b
16 print("Prediction:", y_pred)
```

Output:

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Python + - [ ] [ ] [ ] [ ] [ ]
PS C:\Users\kamul\OneDrive\Desktop\AI ASSISTANT CODING> & C:/Users/kamul/AppData/Local/Programs/Python/Python313/python.exe "c:/Users/kamul/OneDrive/Desktop/AI ASSISTANT CODING/assignment-5.py"
Prediction: 10.0
PS C:\Users\kamul\OneDrive\Desktop\AI ASSISTANT CODING>
```

Responsible Usage Documentation:

- This model assumes **linear relationships** and may not generalize well.
- Predictions may be inaccurate for real-world data.

- Model decisions should be **explainable and interpretable**.
- Bias can exist if training data is limited or skewed.
- AI outputs must always be validated by humans.